

Francis Edward Su

Benediktsson–Karwa Professor of Mathematics · Harvey Mudd College

Department of Mathematics
Harvey Mudd College, 301 Platt Blvd.
Claremont, CA 91711
su@math.hmc.edu · www.math.hmc.edu/su

EDUCATION

HARVARD UNIVERSITY	Ph.D. in Mathematics, 1995 · A.M. in Mathematics, 1992 <i>Methods for Quantifying Rates of Convergence for Random Walks on Groups</i> Advisor: Persi Diaconis
UNIVERSITY OF TEXAS AT AUSTIN	B.S. in Mathematics, 1989 with <i>Highest Honors</i> and <i>Special Honors in Mathematics</i>

ACADEMIC APPOINTMENTS

HARVEY MUDD COLLEGE	Benediktsson–Karwa Professor of Mathematics, 2012–present Professor of Mathematics, 2006–2012 · Interim Chair, 2012 Associate Professor, 2002–2006 Iris and Howard Critchell Assistant Professor, 2001–2002 Assistant Professor, 1996–2001
VISITING POSITIONS	Visiting Researcher, Sydney Mathematical Research Institute, Spring 2024 Chern Professor, Mathematical Sciences Research Institute, Fall 2017 Visiting Associate in Mathematics, Caltech, Fall 2009 Member, Mathematical Sciences Research Institute, Fall 2003 Visiting Assistant Professor, Operations Research, Cornell University, Spring 2000

SELECTED HONORS & AWARDS

2025	Fellow of the American Mathematical Society, for outstanding contributions to the mathematical sciences.
2021	Euler Book Prize (MAA), with Christopher Jackson, for <i>Mathematics for Human Flourishing</i> .
2018	Paul R. Halmos–Lester R. Ford Award, for expository excellence in <i>The American Mathematical Monthly</i> .
2013	Deborah and Franklin Tepper Haimo Award (MAA), for Distinguished College or University Teaching of Mathematics.
2011, '14, '18	Three articles selected for Princeton University Press's annual anthology <i>The Best Writing on Mathematics</i> .
2001	Merten M. Hasse Prize (MAA), for a noteworthy expository paper, by an author under forty.

SELECTED LEADERSHIP & PROFESSIONAL EXPERIENCE

PRESIDENT, MAA	Mathematical Association of America, 2015–2017. Past President 2017–2018; President-Elect 2014–2015. Board of Governors / Directors, 2014–2018.
VICE-PRESIDENT, AMS	American Mathematical Society, 2020–2023. Executive Committee, 2021–2025.
CO-CHAIR, TASK FORCE	Understanding and Documenting the Historical Role of the AMS in Racial Discrimination. Report: <i>Towards a Fully Inclusive Mathematics Profession</i> , 2021.
RESEARCH DIRECTOR	ICERM Research Experiences for Undergraduate Faculty, 2019. MSRI-UP, <i>Geometric Combinatorics</i> , 2015.
CO-ORGANIZER	SMRI special semester, <i>Perspectives on Mathematics</i> , 2024. AMS PREP: <i>Using Your Voice for Influence and Impact</i> . JMM, 2023. MSRI conference: <i>Critical Issues in Mathematics Education</i> , Feb 2018. MSRI semester program, <i>Geometric and Topological Combinatorics</i> , Fall 2017. AMS Mathematics Research Community, Snowbird, 2014.
EDITORIAL BOARDS	MAA FOCUS, 2011–2017, 2022–2027. SIAM Journal on Discrete Mathematics, 2015–2023. The American Mathematical Monthly, 2006–2011.
MENTOR	Mentor to 80+ research and thesis and other students.

SELECTED INVITED ADDRESSES

AMS ARNOLD ROSS LECTURE	<i>Randomness, Geometry, and Privacy</i> . Proof School, San Francisco, 2025.
MAA PÓLYA LECTURER	Invited lectures at six MAA section meetings, 2025–2028.
PME FRAME LECTURE	<i>100 Years of Sperner's Lemma</i> . Pi Mu Epsilon J. Sutherland Frame Lecture, Joint Mathematics Meetings, 2026.
ICME PLENARY PANEL	<i>Does Mathematics Education Effectively Respond to Humanity's Problems?</i> International Congress on Mathematics Education, Sydney, 2024.
IRIS M. CARL EQUITY ADDRESS	<i>Finite Disappointment, Infinite Hope</i> . NCTM Annual Meeting, 2021.
PHI BETA KAPPA LECTURER	One of 15 Visiting Scholars across all disciplines giving PBK invited lectures on campuses across the US, 2019–2021.
NCTM CLOSING PLENARY	<i>Mathematics for Human Flourishing</i> . NCTM Annual Meeting, 2018.
MAA PRESIDENTIAL ADDRESS	<i>Mathematics for Human Flourishing</i> . Joint Mathematics Meetings, Atlanta, 2017.
MAA INVITED ADDRESS	<i>Fair Division</i> . 2016 AMS–MAA Joint Mathematics Meetings, Seattle.
HAIMO AWARD LECTURE	<i>The Lesson of Grace in Teaching</i> . Joint Math Meetings, San Diego, 2013.
MAA LEITZEL LECTURE	<i>Teaching Research: Encouraging Discoveries</i> . MathFest, 2006.

BOOKS & MAJOR MEDIA

Mathematics for Human Flourishing, with reflections by Christopher Jackson. Yale University Press, 2020.

Winner, 2021 Euler Book Prize. Finalist, Phi Beta Kappa Science Award. 28,000+ copies sold.

Translated into 9 languages: Chinese (traditional & simplified), Japanese, Korean, Spanish, Italian, Turkish, Romanian, Greek.

Topology Through Inquiry, with Michael Starbird. AMS/MAA Press, 2019.

Mastering Linear Algebra, video series (24 lectures), The Great Courses, 2019. Surveys the topics of a first semester linear algebra course with modern applications.

YouTube Course on Real Analysis, 26 lectures, 2010-2012. Viewed 2.05M+ times.

Math Fun Facts website, 1999–present. Archive of 200+ mathematical facts; 1M+ visits per year.

SELECTED PUBLICATIONS

RESEARCH PUBLICATIONS

In pure mathematics the authorship convention is alphabetical; undergraduate co-authors are starred ().*

C. Cannone, F. Su, S. Vadhan. The Randomness Geometry of Differential Privacy. *16th Innovations in Theoretical Computer Science Conference (ITCS 2025)*, LIPIcs vol. 325, pp. 27:1–27:21, 2025.

R. Alvarado, M. Averett, B. Gaines, C. Jackson, M. L. Karker, M. A. Marciniak, F. E. Su, S. Walker. The Game of Cycles. *The American Mathematical Monthly* 128 (2021), 868–887.

K. Nyman, F. E. Su, S. Zerbib. Fair division with multiple pieces. *Discrete Applied Mathematics* 283 (2020), 115–122.

F. Meunier, F. E. Su. Multilabeled versions of Sperner’s and Fan’s lemmas and applications. *SIAM Journal on Applied Algebra and Geometry* 3 (2019), 391–411.

F. E. Su, S. Zerbib. Piercing numbers in approval voting. *Mathematical Social Sciences* 101 (2019), 65–71.

K. Nyman, F. E. Su. A Borsuk–Ulam equivalent that implies Sperner’s lemma. *The American Mathematical Monthly* 120 (2013), 346–354.

C. J. Haake, A. Kashiwada*, F. E. Su. The Shapley value of phylogenetic trees. *J. Math. Biol.* 56 (2008), 479–497.

J. A. De Loera, E. Peterson*, F. E. Su. A polytopal generalization of Sperner’s lemma. *J. Combin. Theory Ser. A* 100 (2002), 1–26.

A. L. Gibbs, F. E. Su. On choosing and bounding probability metrics. *Internat. Statist. Rev.* 70 (2002), no. 3, 419–435.

F. E. Su. Rental harmony: Sperner’s lemma in fair division. *The American Mathematical Monthly* 106 (1999), 930–942. **Winner, 2001 MAA Hasse Prize.**

F. E. Su. Convergence of random walks on the circle generated by an irrational rotation. *Trans. Amer. Math. Soc.* 350 (1998), 3717–3741.

EXPOSITORY & PROFESSIONAL WRITING

- T. Inniss, J. Lewis, I. Mitrea, K. Okoudjou, A. Salerno, F. Su, D. Thurston. *Towards a Fully Inclusive Mathematics Profession*. Report of the AMS Task Force. American Mathematical Society, 2021.
- F. E. Su. “Should I Quit Mathematics?” In *Living Proof: Stories of Resilience Along the Mathematical Journey*. AMS, 2019. Reprinted in *Notices of the AMS*, Oct. 2019.
- NCTM Writing Team. *Catalyzing Change in High School Mathematics: Initiating Critical Conversations*. NCTM, 2018.
- F. E. Su. Mathematics for Human Flourishing. *The American Mathematical Monthly* 124 (2017), 483–493. **Winner, 2018 Halmos–Ford Award**. Also in *The Best Writing on Mathematics 2018* (Princeton Univ. Press).
- F. E. Su. Some guidelines for good mathematical writing. *MAA FOCUS*, Aug./Sep. 2015, 20–22.
- F. E. Su. Teaching research: encouraging discoveries. *The American Mathematical Monthly* 117 (2010), 159–169. Also in *The Best Writing on Mathematics 2011* (Princeton Univ. Press).

PUBLIC & POPULAR WRITING

- F. Su, A. C. Hanbury. “Math Isn’t Good for You. It’s Good for Us.” *Common Good* magazine, Dec. 2023.
- J. A. De Loera, F. Su. “Calculus isn’t the only option. Let’s broaden and update the current math curriculum.” *The Sacramento Bee*, June 5, 2022.
- F. Su. “Can Mathematics Be Spiritual?” John Templeton Foundation / Big Think, Feb. 2022.
- F. E. Su. “Solve this math problem: the gender gap.” *Los Angeles Times*, Aug. 25, 2014.
- F. E. Su. The Lesson of Grace in Teaching. In *The Best Writing on Mathematics 2014* (Princeton Univ. Press).

SELECTED GRANTS

- | | |
|-----------|---|
| 2010–2014 | NSF Research Award DMS-1002938, \$205,668. RUI: <i>Triangulations, Set Intersections, Fair Division, and Voting</i> . |
| 2007–2010 | NSF Research Award DMS-0701308, \$114,468. RUI: <i>Combinatorial Fixed Point Theorems, Polytopes, and Preference Sets</i> . |
| 2003–2006 | NSF Research Award DMS-0301129, \$146,640. RUI: <i>Combinatorial Fixed Point Theorems, Polytopes, and Fair Division</i> . |

COMPLETE CURRICULUM VITAE AVAILABLE UPON REQUEST. Full publication list, complete talks, grants, professional and college service, and teaching record.